

Contents

1 - Mint (create the wallet) 2

2 - Spend (base M-of-N + delegated authorizations) 3

3 - Update withdrawals (per-UTxO) 4

4 - Close (admin burns the NFT) 5

1 - Mint (create the wallet)

The wallet is created by minting its identifying NFT — one token under the script’s own policy — into a fresh wallet UTxO at the wallet address. The admin must authorize the mint.

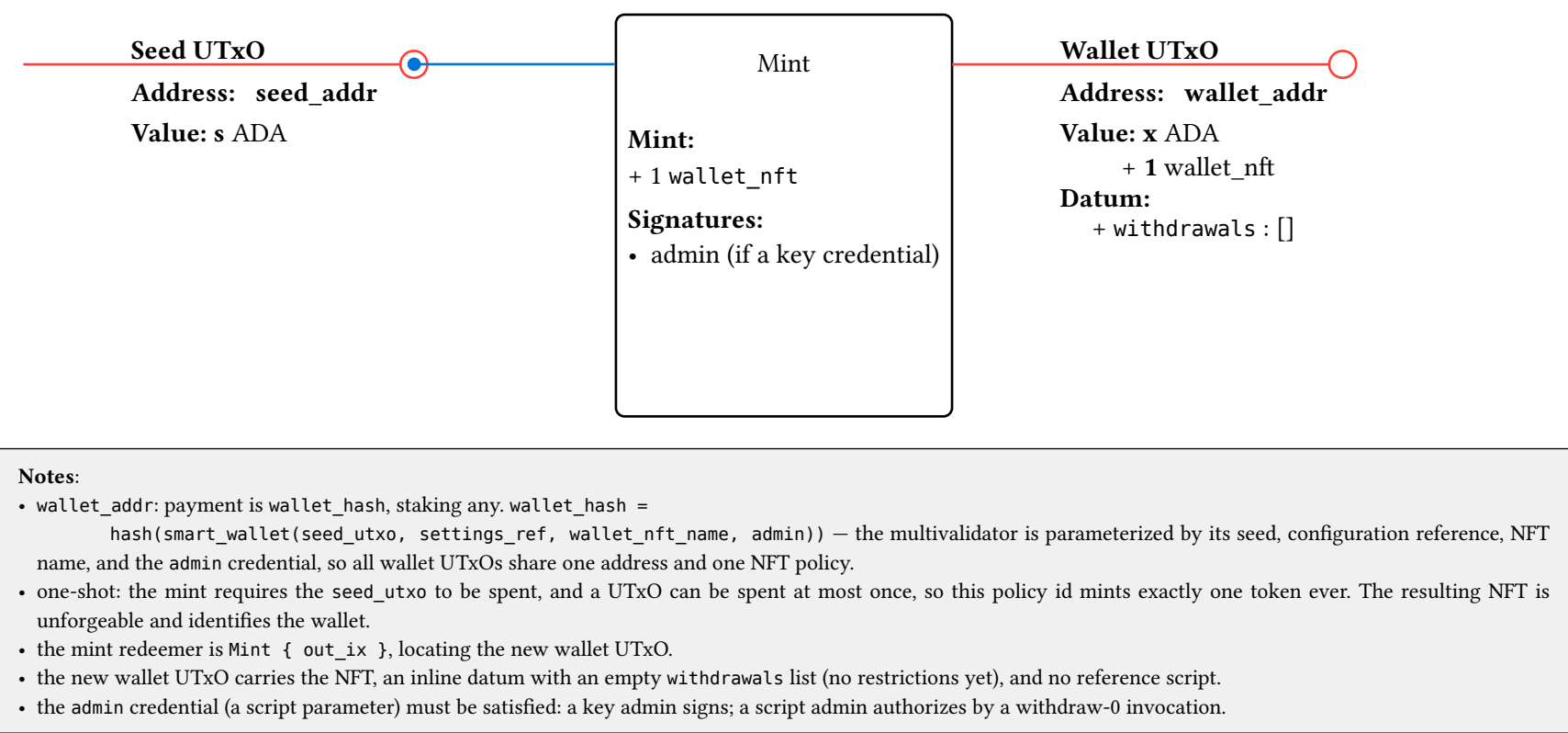
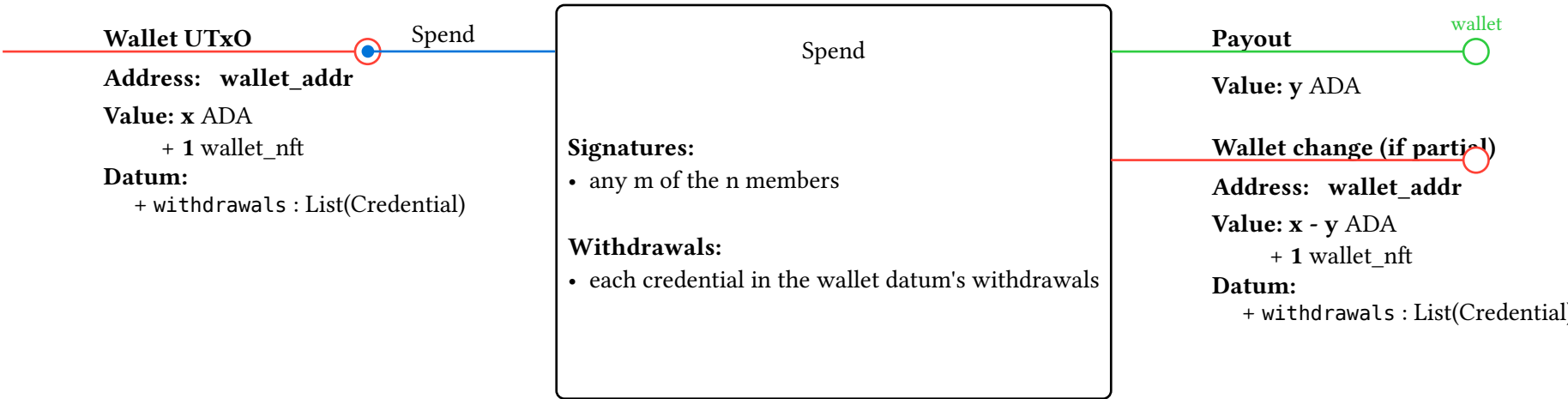


Figure 1: Mint (create the wallet)

2 - Spend (base M-of-N + delegated authorizations)

The wallet pays out. The spending script enforces its base M-of-N signature floor and forwards every other check to the withdrawal scripts named in the wallet UTxO's datum; the single spend redeemer asserts their conjunction.



Notes:

- `wallet_addr`: payment is `wallet_hash`, staking any. A wallet UTxO holds its identifying NFT plus ada and arbitrary native assets.
- `wallet_hash = hash(smart_wallet(seed_utxo, settings_ref, wallet_nft_name, admin))`: the spending script is parameterized by its seed, configuration reference, NFT name, and the `admin` credential, so all wallet UTxOs share one address. Externalizing the members/threshold config lets it change without redeploying the script.
- validity: the spend requires the wallet UTxO to carry its NFT (1 `wallet_nft` under `wallet_hash`) — only NFT-bearing UTxOs are valid smart wallets.
- base M-of-N floor: the wallet's configuration supplies members and threshold, and the spend requires at least threshold of members present in `extra_signatories`. A member may itself be a script credential, so the floor can be another multisig, a DAO, and so on. This design is agnostic of how that configuration is sourced.
- delegated authorizations: every `Credential` in the wallet UTxO's `withdrawals` datum field is a `withdraw-0` staking script that must run in this transaction. The Spend redeemer asserts the conjunction — the M-of-N floor **and** all delegated checks — so no restriction can be silently omitted. The list is per-UTxO, so different wallet UTxOs may carry different restriction sets.
- the delegated scripts are the extension point: spending limits, time locks, recipient allow-lists, oracle attestations, ... each is an independent withdrawal script observed here, none known to the spending script.
- composability: the spend asserts only its own inputs and that the required withdrawals ran — never the total input or output counts, nor unrelated UTxOs. Any partial change keeps the NFT so the wallet stays valid.

Figure 2: Spend (base M-of-N + delegated authorizations)

3 - Update withdrawals (per-UTxO)

An authorized admin credential — a parameter of the spending script — rewrites a wallet UTxO’s delegated `withdrawals` list in place. Funds are untouched: same address, same value — only the restriction set changes.

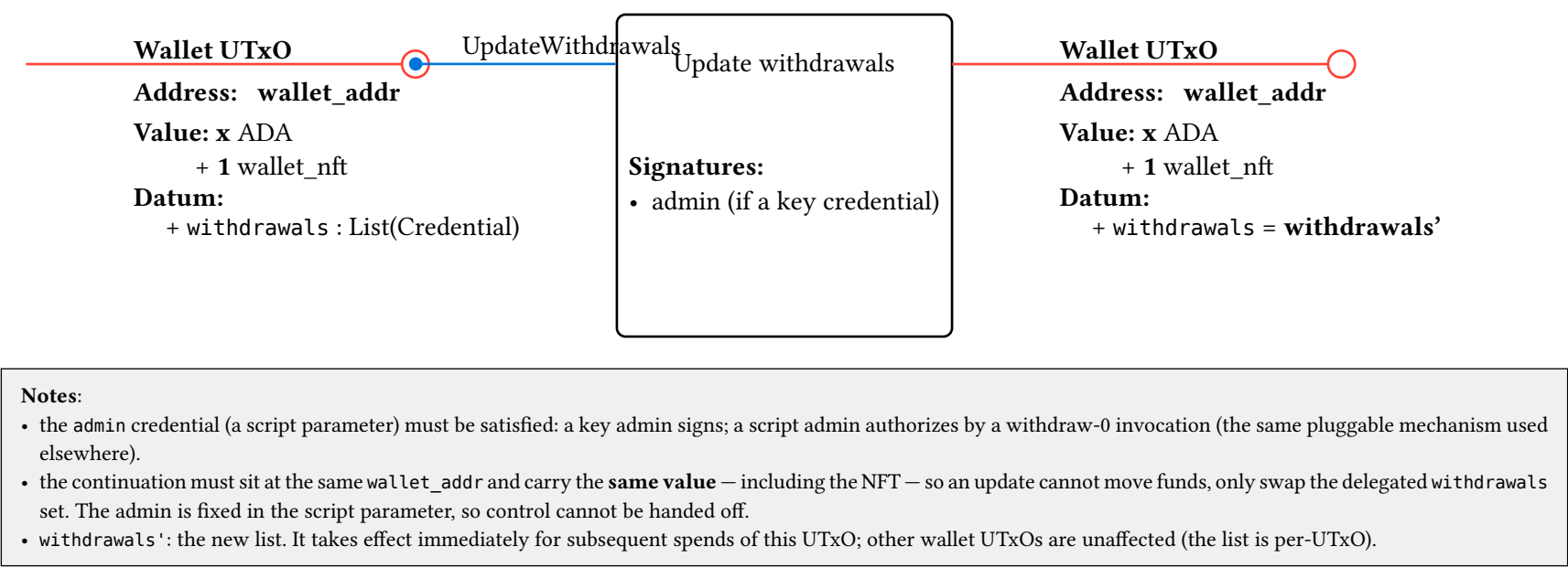


Figure 3: Update withdrawals (per-UTxO)

4 - Close (admin burns the NFT)

The admin closes the wallet: the wallet UTxO is spent, releasing whatever funds remain, and the identifying NFT is burned. The Close spend redeemer requires the admin and the burn; the Burn mint redeemer permits it.

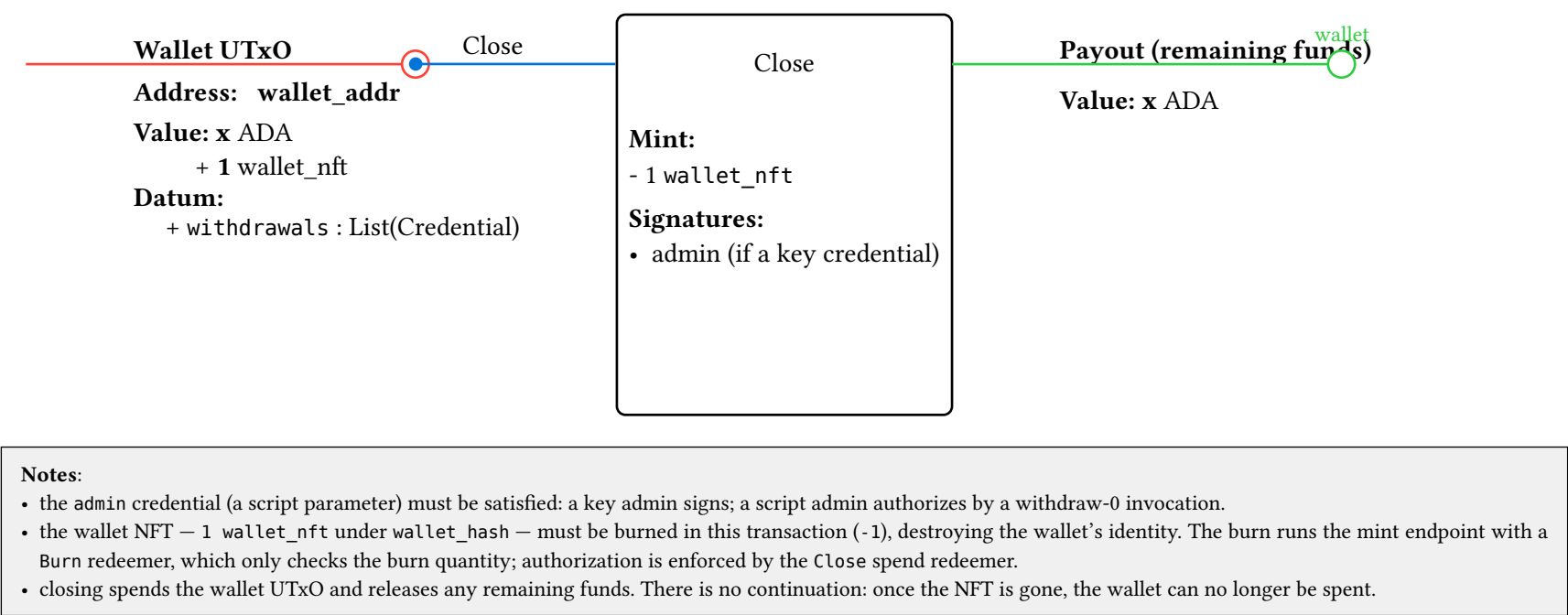


Figure 4: Close (admin burns the NFT)